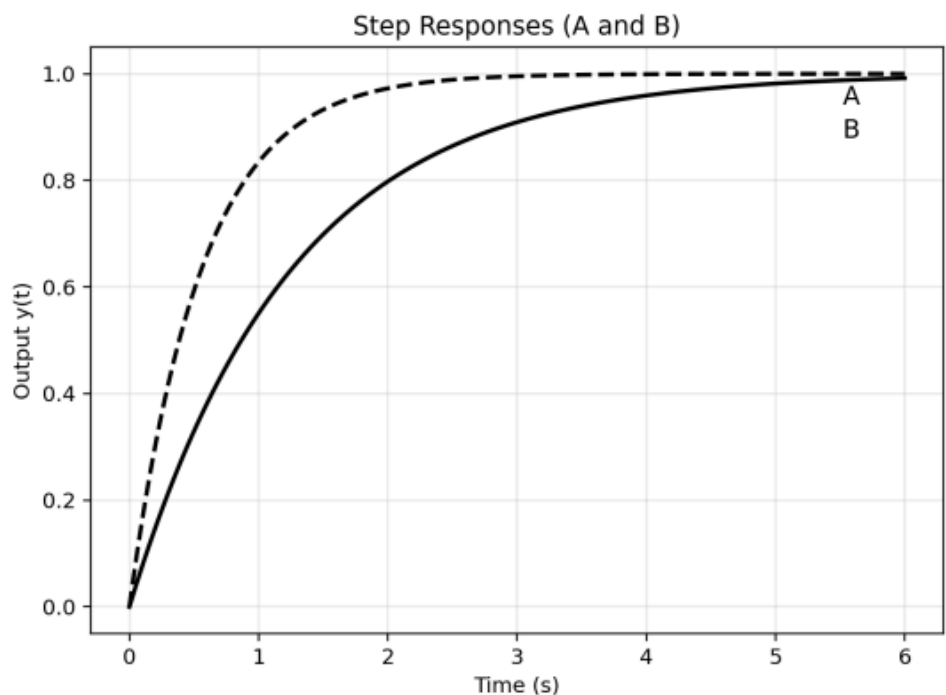
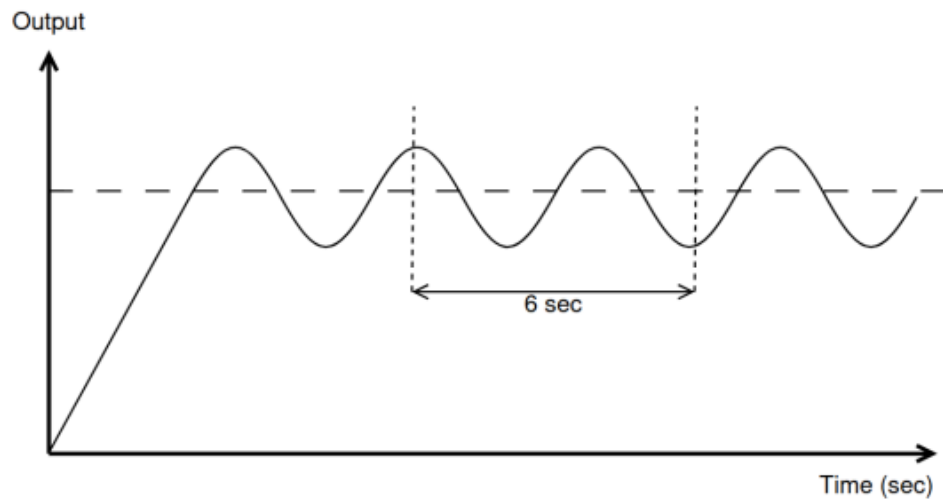


Question 1: See the diagram below and write the relations (e.g., >, <, =, faster/slower) in the blanks.



Rise time (T_r): A _____ B
Settling time (T_s): A _____ B

Question 2: Consider the following sustained oscillation figure.



Question 2 (continued):

For this figure the ultimate gain was found as **K**. After finding the ultimate gain K, your CSE461 course instructor uses only P controller. When you curious mind ask the instructor what is the value of **K_p** using the Ziegler–Nichols method, he gave you a hint in the below

Hint: **K_p = 0.611K – 1**

Write the values:

Ultimate Gain, K =

Ultimate Period, P =

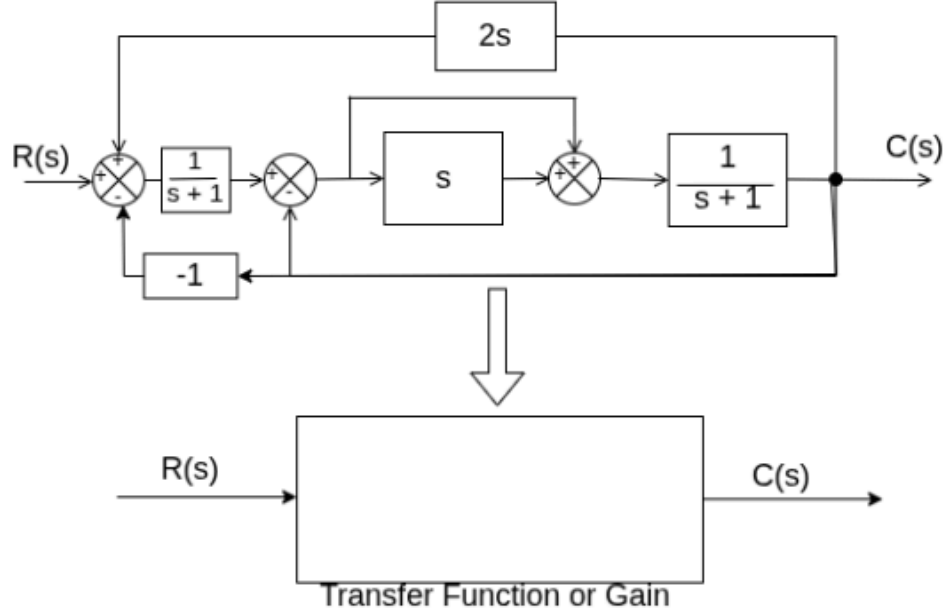
Now you want to use a PID controller for the same system. Therefore what will be your **K_p**, **K_i** and **K_d** values using Ziegler–Nichols method?

K_p =

K_i =

K_d =

Question 3: Block diagram reduction.



(a) Write down the Transfer Function or Gain of the whole system in the given box.

(b) If the Transfer Function or Gain = ∞, what will be the value of s?

s =

Question 4: PID control for an AC system.

An air conditioning (AC) system is currently set to 18°C. You feel too cold, so you want to increase the temperature to 25°C using a PID controller. Given K_p = 0.1, K_i = 0.02, K_d = 0.01. The error at step 0 is e₀ = 0. Please write down the next 5 steps in the table below.

Step	Error	PID value	Current Temperature
1			
2			
3			
4			
5			